



Site-to-Site IPSec VPN Security Lab

IKE Negotiation, IPSec Encryption, NAT Exemption,
Traffic Filtering, Attack Testing and Security Validation

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Cisco IOSv / GNS3

Secure communication between two remote networks

The objective was to establish a **site-to-site IPSec VPN** between two networks connected through an untrusted WAN.

The project also examined how NAT, ACL placement, control-plane traffic and management access affect the security of an encrypted tunnel.

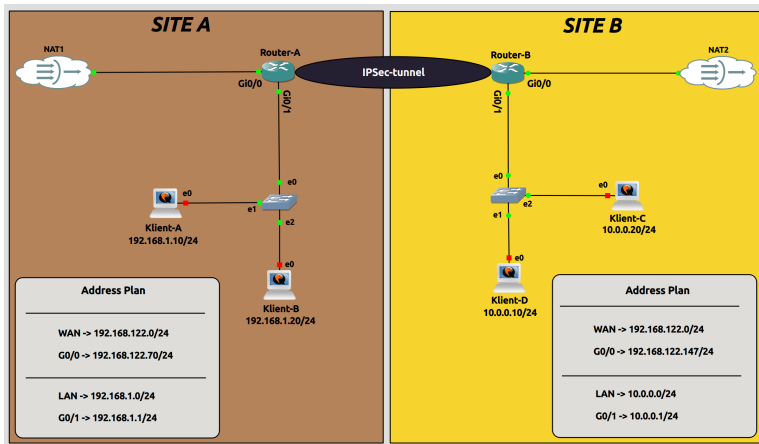
Main objectives

- ▶ Build and verify an IPSec tunnel
- ▶ Protect Site A ↔ Site B traffic
- ▶ Implement NAT exemption
- ▶ Filter unwanted WAN traffic
- ▶ Test Nmap traffic through and outside the tunnel
- ▶ Mitigate ICMP flooding
- ▶ Restrict SSH management access

Technologies

Cisco IOSv
IPSec / ESP
IKE / ISAKMP
AES-256 / SHA-256
Extended ACLs
NAT / PAT
Control Plane Policing
Nmap / hping3
Wireshark
GNS3

Lab Topology



Site A

LAN

192.168.1.0/24

Router A LAN

192.168.1.1

Router A WAN

192.168.122.70

Site B

LAN

10.0.0.0/24

Router B LAN

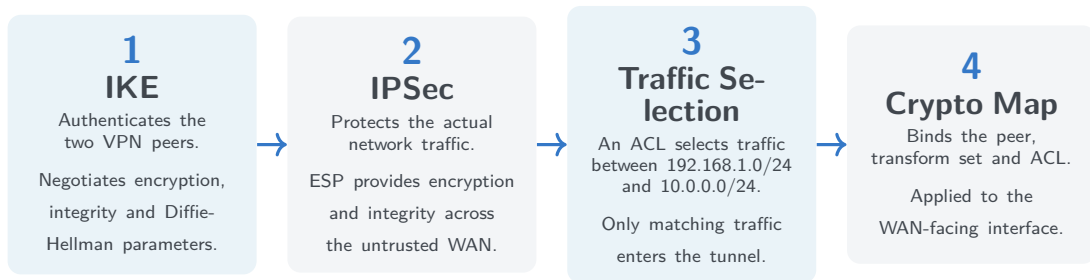
10.0.0.1

Router B WAN

192.168.122.147

Actual GNS3 environment
used for the project.

How the IPSec Tunnel Is Built



IKE Policy — Secure Negotiation

Router A configuration

```
crypto isakmp policy 1
  encryption aes 256
  hash sha256
  authentication pre-share
  group 14

crypto isakmp key <REDACTED-PSK> address
192.168.122.147
```

PSK intentionally redacted. Router B uses 192.168.122.70 as its peer.

Security parameters

- ▶ **AES-256**
Encryption
- ▶ **SHA-256**
IKE integrity
- ▶ **DH Group 14**
Key exchange
- ▶ **PSK**
Peer authentication

Technology context: IKEv1 with a pre-shared key was used in this lab. A modern production design would normally consider IKEv2 and certificate-based authentication.

IPSec Protection and Traffic Selection

Router A

```
crypto ipsec transform-set Site2Site-TS esp-aes 256  
esp-sha-hmac  
crypto ipsec transform-set Site2Site-TS mode tunnel  
  
ip access-list extended Site2Site-Traffic  
permit ip 192.168.1.0 0.0.0.255 10.0.0.0 0.0.0.255
```

Router B uses the mirrored source and destination networks.

ESP-AES-256

Encrypts packets travelling through the tunnel.

ESP-SHA-HMAC

Uses HMAC-SHA-1 for integrity and authentication.

Traffic ACL

Defines which packets should enter the VPN.

Technology context: esp-sha-hmac uses HMAC-SHA-1. This is the algorithm implemented in the lab; stronger integrity algorithms or authenticated-encryption modes would be preferred in a modern production design.

Crypto Map — Binding the VPN Components

```
crypto map Site2Site_MAP 1 ipsec-isakmp
set peer 192.168.122.147
set transform-set Site2Site-TS
match address Site2Site-Traffic

interface GigabitEthernet0/0
crypto map Site2Site_MAP
```

Crypto map

Combines:

- ▶ Remote VPN peer
- ▶ IPSec transform set
- ▶ Interesting-traffic ACL

The complete policy is then applied to the WAN-facing interface.

Design context: This lab uses a classic policy-based IPSec VPN with crypto maps. Modern deployments may instead use route-based IPSec with Virtual Tunnel Interfaces (VTIs).

NAT Exemption — Preserving VPN Traffic

```
ip access-list extended NAT_EXCEPTION
deny ip 192.168.1.0 0.0.0.255 10.0.0.0 0.0.0.255
permit ip 192.168.1.0 0.0.0.255 any

ip nat inside source list NAT_EXCEPTION interface
GigabitEthernet0/0 overload
```

VPN traffic

192.168.1.0/24



10.0.0.0/24

NO NAT

Internet traffic

192.168.1.0/24



Internet

NAT/PAT

Actual troubleshooting issue: VPN traffic was initially NATed before encryption and therefore failed to match the IPSec traffic-selection ACL correctly.

Testing & Validation

Tunnel verification, packet inspection, Nmap scanning, ICMP flood protection and SSH hardening

Proof of Tunnel Establishment

Actual Router A verification output

```
Router-A# show crypto isakmp sa

dst          src          state    status
192.168.122.147 192.168.122.70 QM_IDLE  ACTIVE

Router-A# show crypto ipsec sa

local ident: 192.168.1.0/24
remote ident: 10.0.0.0/24
current_peer: 192.168.122.147

#pkts encaps: 12
#pkts encrypt: 12
#pkts decaps: 12
#pkts decrypt: 12
```

What this proves

- ▶ IKE peer is active
- ▶ Negotiation succeeded
- ▶ Correct networks are protected
- ▶ Outbound packets are encrypted
- ▶ Return packets are decrypted

Connectivity alone does not prove encryption. The IPSec counters provide the evidence.

Proof of Encrypted Traffic

```
Router-A# show crypto ipsec sa | include pkts

#pkts encaps: 20
#pkts encrypt: 20
#pkts decaps: 20
#pkts decrypt: 20

[additional traffic generated]

Router-A# show crypto ipsec sa | include pkts

#pkts encaps: 41
#pkts encrypt: 41
#pkts decaps: 41
#pkts decrypt: 41
```

Observed result

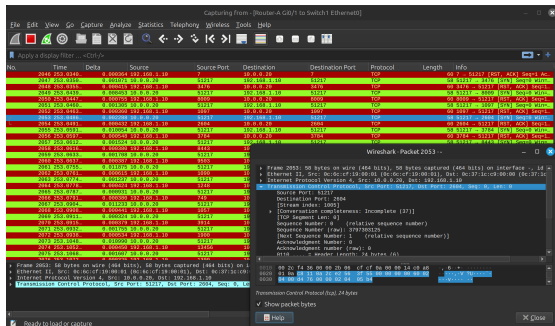
After additional traffic:

- ▶ Encapsulation increased
- ▶ Encryption increased
- ▶ Decapsulation increased
- ▶ Decryption increased

Counters changed from **20** to **41** packets.

This confirms bidirectional encrypted traffic.

Wireshark Verification — Internal Network



What Wireshark sees

The internal capture exposes the original **TCP traffic** between 10.0.0.20 and 192.168.1.10.

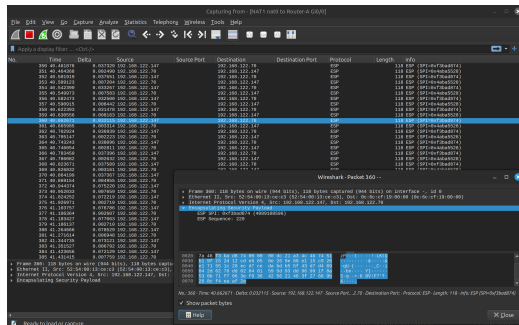
Source/destination addresses, TCP ports, flags and header information remain visible.

Analysis

This capture was taken on the protected **internal** side of the VPN endpoint.

At this point the packet is available in its original form before or after IPsec processing, depending on traffic direction.

Wireshark Verification – ESP on the WAN



What Wireshark sees

The WAN capture identifies packets as **Encapsulating Security Payload (ESP)**.

The visible IP addresses are the VPN peers:

192.168.122.147 and **192.168.122.70**.

What this proves

Wireshark sees the ESP header, SPI and sequence information, but not the original TCP packet in clear form.

This is packet-level evidence that IPsec is protecting the communication across the WAN.

End-to-End Connectivity Test

```
Testing host > ping 10.0.0.20  
  
64 bytes from 10.0.0.20: icmp_seq=1  
64 bytes from 10.0.0.20: icmp_seq=2  
64 bytes from 10.0.0.20: icmp_seq=3  
64 bytes from 10.0.0.20: icmp_seq=4  
  
4 packets transmitted  
4 received  
0% packet loss
```

Local Linux username and hostname removed.

Result

The Site A client successfully reached the Site B network.

Combined with the IPSec counters:

- ▶ Routing works
- ▶ VPN ACL matches
- ▶ Traffic is encrypted
- ▶ Return traffic is decrypted

WAN Filtering — Actual ACL

```
Extended IP access list WAN_INBOUND

10 permit udp host 192.168.122.147
   host 192.168.122.70 eq isakmp          (16 matches)

20 permit esp host 192.168.122.147
   host 192.168.122.70                  (17539 matches)

30 permit tcp any any established

70 deny tcp 10.0.0.0 0.0.0.255
   host 192.168.1.1 log

80 permit icmp 10.0.0.0 0.0.0.255
   host 192.168.1.1 echo

90 deny ip any any log                  (2492 matches)
```

Purpose

- ▶ Allows IKE
- ▶ Allows ESP
- ▶ Allows established TCP
- ▶ Restricts unwanted traffic
- ▶ Logs denied packets

The counters show that the ACL matched real lab traffic.

Nmap Test 1 — Scan from the WAN

Sanitized actual scan output

```
Testing host > sudo nmap 192.168.122.70

Nmap scan report for 192.168.122.70
Host is up.

All 1000 scanned ports are in ignored states.

Not shown:
1000 filtered tcp ports (admin-prohibited)

Nmap done:
1 IP address (1 host up)
```

Username, hostname, MAC address and timestamps omitted.

Observed result

The scan targeted Router A directly from the WAN side.

1000 TCP ports were filtered.

This confirms that the WAN-facing ACL restricted unsolicited scan traffic.

Nmap Test 2 — Scan Through the IPSec Tunnel

```
Site B test host >  
sudo nmap -sS -Pn 192.168.1.1  
  
Nmap scan report for 192.168.1.1  
Host is up.  
  
All 1000 scanned ports are in ignored states.  
  
Not shown:  
1000 closed tcp ports (reset)  
  
Nmap done:  
1 IP address (1 host up)
```

Username, hostname and timestamps removed.

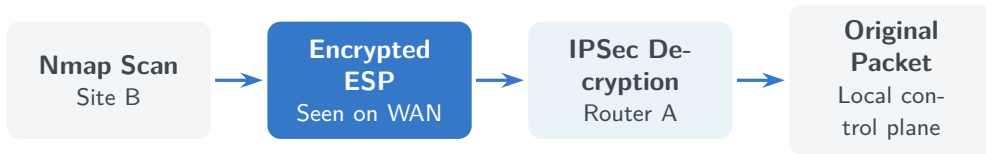
Unexpected result

The WAN ACL did **not** filter this scan in the same way.

Why?

1. WAN sees encrypted ESP
2. Router decrypts ESP
3. Original packets appear later
4. Traffic to 192.168.1.1 is processed locally

Why the VPN Scan Behaved Differently



Key troubleshooting lesson: security filtering must account for where packets are inspected **before and after IPSec decryption**.

ICMP Flood Protection — Control Plane Policing

```
ip access-list extended ICMP_ECHO
permit icmp any any echo

class-map match-all ICMP_CLASS
match access-group name ICMP_ECHO

policy-map ICMP_LIMIT
class ICMP_CLASS
police 8000 conform-action transmit exceed-action
drop

control-plane
service-policy input ICMP_LIMIT
```

Why this was required

An ACL can identify ICMP traffic but cannot rate-limit it alone.

CoPP was used to:

- ▶ Classify ICMP echo traffic
- ▶ Permit conforming traffic
- ▶ Drop excess traffic

IOSv limitation:

minimum supported policing rate was **8000 bps**.

ICMP Flood Test — Before and After

Traffic generation

```
Site B test host >  
sudo hping3 --icmp --flood 192.168.1.1  
  
HPING 192.168.1.1: icmp mode set  
hping in flood mode  
  
1364994 packets transmitted  
0 packets received
```

Control-plane counters

```
Before:  
0 matched packets  
0 exceeded packets  
  
After:  
2951 matched packets  
154 conformed packets  
2797 exceeded packets -> drop
```

Observed result

2,951 packets matched.

154 packets conformed.

2,797 packets exceeded the rate and were dropped.

This provides direct evidence that the control-plane policy was actively limiting excessive ICMP traffic.

SSH Hardening — Limiting Management Exposure

Implemented VTY restriction

```
access-list 10 deny 10.0.0.0 0.0.0.255
access-list 10 permit 192.168.1.0 0.0.0.255

line vty 0 4
access-class 10 in
login local
transport input ssh
```

```
Router-A# show access-list 10

10 deny   10.0.0.0 0.0.0.255 (3 matches)
20 permit 192.168.1.0 0.0.0.255
```

Security goal

A compromised host at Site B should not automatically gain management access to Router A.

Site B

10.0.0.0/24

SSH denied

Site A

192.168.1.0/24

SSH permitted

The ACL counter confirmed blocked SSH attempts.

SSH Protection — Nmap Proof

Before restriction

```
192.168.1.1      sudo nmap -sS -Pn 192.168.1.1

Nmap scan report for

Host is up.

Not shown:
999 closed tcp ports

PORT      STATE SERVICE
22/tcp    open  ssh
```

After VTY ACL

```
192.168.1.1      sudo nmap -sS -Pn 192.168.1.1

Nmap scan report for

Host is up.

All 1000 scanned ports
are in ignored states.

Not shown:
1000 closed tcp ports
```

Observed result: TCP/22 no longer appeared as an open SSH service from the Site B test host after the VTY restriction.

Troubleshooting — What Actually Went Wrong

1. NAT Exemption

Problem

VPN traffic was NATed before encryption.

Effect

Traffic no longer matched the IPsec ACL.

Fix

Exclude VPN traffic from NAT.

2. Wrong Peer

Problem

The local address was initially entered as the peer.

Effect

The tunnel failed to establish correctly.

Fix

Use the remote WAN IP.

3. ACL Placement

Problem

WAN ACL did not stop the VPN Nmap scan.

Cause

WAN sees ESP; original traffic appears after decryption.

Lesson

Packet-processing order matters.

Technology Context — Lab vs Modern Production

Classic implementation, modern security awareness

The VPN configuration is valid and demonstrated the core IPsec concepts successfully. Some components represent a classic design compared with newer production approaches.

Implemented in this lab

IKEv1 + pre-shared key

Peer authentication and key negotiation.

Classic crypto-map VPN

Policy-based IPsec implementation.

ESP-SHA-HMAC

HMAC-SHA-1 integrity protection.

Modern production consideration

IKEv2 + certificates

Stronger and more scalable peer authentication.

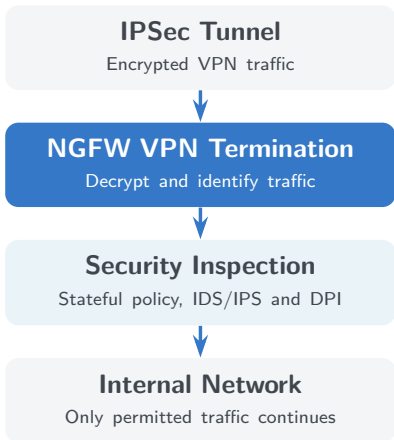
Route-based VPN / VTI

Greater routing flexibility than classic crypto maps.

Stronger integrity / AEAD

Modern cryptographic protection where supported.

Why Use a Next-Generation Firewall?



More than basic ACL filtering

After terminating and decrypting the IPSec tunnel, an NGFW can apply richer security controls:

- ▶ Stateful firewall policy
- ▶ Application-aware filtering
- ▶ IDS / IPS inspection
- ▶ Deep Packet Inspection
- ▶ Threat and malware detection
- ▶ Centralized logging
- ▶ More granular segmentation

Application-layer encryption can still limit payload visibility unless additional inspection is available and appropriate.

Production Security Considerations

Limitations of the lab design

The Cisco routers primarily provide encryption, routing and ACL-based filtering.

A compromised remote host may still attempt attacks through the trusted VPN tunnel.

- ▶ IKEv1 / PSK authentication
- ▶ SHA-1-based ESP integrity
- ▶ Policy-based crypto maps
- ▶ Limited application inspection
- ▶ Limited post-decryption visibility

Modern production direction

- ▷ IKEv2
- ▷ Certificate-based authentication
- ▷ Stronger integrity / AEAD
- ▷ Route-based VPN using VTIs
- ▷ NGFW as the VPN security gateway
- ▷ IDS/IPS and DPI after decryption
- ▷ Centralized monitoring and logging

Encryption protects the transport path.

It does not automatically make traffic inside the VPN trustworthy.

Project Summary

A tested and hardened site-to-site IPSec deployment

Implemented

- ▶ IKEv1 with AES-256, SHA-256 and DH14
- ▶ IPSec ESP tunnel protection
- ▶ Crypto maps and traffic-selection ACLs
- ▶ NAT exemption
- ▶ WAN and SSH management hardening

Validated

- ▶ Active IKE/IPSec Security Associations
- ▶ Encryption/decryption counters
- ▶ Wireshark proof of ESP across the WAN
- ▶ WAN and VPN Nmap testing
- ▶ ICMP flood mitigation with CoPP
- ▶ Restricted SSH management access

Result

The project went beyond simply creating a VPN tunnel.

It demonstrated **configuration, verification, packet inspection, attack testing, troubleshooting and hardening.**

Cisco counters and Wireshark independently showed that traffic was protected by ESP across the WAN.

The lab also identified how the classic design could be modernized without claiming those improvements were implemented.